

Vitamin B12

Fact Sheet for Health Professionals

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Introduction

Vitamin B12 is a water-soluble vitamin that is naturally present in some foods, added to others, and available as a dietary supplement and a prescription medication. Because vitamin B12 contains the mineral cobalt, compounds with vitamin B12 activity are collectively called cobalamins [1]. Methylcobalamin and 5-deoxyadenosylcobalamin are the metabolically active forms of vitamin B12. However, two others forms, hydroxycobalamin and cyanocobalamin, become biologically active after they are converted to methylcobalamin or 5-deoxyadenosylcobalamin [1-3].

Vitamin B12 is required for the development, myelination, and function of the central nervous system; healthy red blood cell formation; and DNA synthesis [1,4,5]. Vitamin B12 functions as a cofactor for two enzymes, methionine synthase and L-methylmalonyl-CoA mutase [1-3,5]. Methionine synthase catalyzes the conversion of homocysteine to the essential amino acid methionine [1,2]. Methionine is required for the formation of S-adenosylmethionine, a universal methyl donor for almost 100 different substrates, including DNA, RNA, proteins, and lipids [3,5]. L-methylmalonyl-CoA mutase converts L-methylmalonyl-CoA to succinyl-CoA in the metabolism of propionate, a short-chain fatty acid [2].

Vitamin B12 is bound to protein in food and must be released before it is absorbed [5]. The process starts in the mouth when food is mixed with saliva. The freed vitamin B12 then binds with haptocorrin, a cobalamin-binding protein in the saliva. More vitamin B12 is released from its food matrix by the activity of hydrochloric acid and gastric protease in the stomach, where it then binds to haptocorrin [1]. In the duodenum, digestive enzymes free the vitamin B12 from haptocorrin, and this freed vitamin B12 combines with intrinsic factor, a transport and delivery binding protein secreted by the stomach's parietal cells. The resulting complex is absorbed in the distal ileum by receptor-mediated endocytosis [1,5]. If vitamin B12 is added to fortified foods and dietary supplements, it is already in free form and therefore does not require the separation step.

Assessing Vitamin B12 Status

Serum or plasma vitamin B12 levels are typically used to assess a person's vitamin B12 status, with values lower than 200 or 250 pg/mL generally being considered subnormal. The most sensitive marker of vitamin B12 status is a vitamin B12-associated metabolite called methylmalonic acid, and serum levels of this metabolite can be used to confirm a diagnosis of vitamin B12 deficiency if a patient's serum vitamin B12 level is between 150 and 399 pg/mL.

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Recommended Intakes

The Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine has established Recommended Dietary Allowances and Adequate Intakes for vitamin B12. These values range from 2.4 to 2.8 mcg for adults and from 0.4 to 2.8 mcg for infants, children, and adolescents, depending on age.

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Sources of Vitamin B12

Food

Vitamin B12 is found in foods of animal origin. It is not naturally present in plant foods, but some plant foods are fortified with vitamin B12, including many

breakfast cereals and nutritional yeasts. Breast milk also contains a small amount of vitamin B12. The bioavailability of vitamin B12 from food varies by vitamin B12 dose and by the type of food source.

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Dietary supplements

Vitamin B12 is present in many dietary supplements, often as cyanocobalamin, but other forms may also be used. Supplements that only contain vitamin B12 typically have the highest doses of the vitamin, while multivitamin/mineral supplements and supplements that combine vitamin B12 with other B-complex vitamins usually contain lower doses.

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Prescription medications

Certain forms of vitamin B12 are available as prescription medications. These are generally used to treat vitamin B12 deficiency.

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Vitamin B12 Intakes and Status

According to data from the National Health and Nutrition Examination Survey, most people in the United States have sufficient intakes of vitamin B12.

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Vitamin B12 Deficiency

Even people who have sufficient intakes of vitamin B12 may have vitamin B12 deficiency, due to factors such as difficulty absorbing the vitamin from food, lack of intrinsic factor due to pernicious anemia, or the prolonged use of certain medications. Vitamin B12 deficiency can cause a number of symptoms, including fatigue, neurological changes, megaloblastic anemia, glossitis of the tongue, palpitations, and low counts of white and red blood cells. The symptoms of vitamin B12 deficiency can take several years to appear.

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Groups at Risk of Vitamin B12 Inadequacy

Certain groups of people are more likely than others to be vitamin B12 deficient. These include older adults, people with pernicious anemia or gastrointestinal disorders, and those who have had gastrointestinal surgery. Other groups that are at risk of vitamin B12 deficiency include people who follow vegetarian diets and the infants of women who follow vegan diets.

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Vitamin B12 and Health

This section focuses on areas of health in which vitamin B12 might be involved: cancer, cardiovascular disease (CVD), dementia and cognitive function, and energy and endurance.

Cancer

Some evidence from observational studies and clinical trials suggests that higher intakes or blood concentrations of vitamin B12 may increase the risk of cancer. However, other evidence contradicts those findings, showing a link between lower intakes or concentrations of vitamin B12 and an increased cancer risk or no link at all. More research is needed to clarify the association between high or low intakes of vitamin B12 and the risk of cancer.

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Cardiovascular disease

Because vitamin B12 plays a role in homocysteine metabolism, researchers have investigated whether vitamin B12 supplementation can reduce the risk of cardiovascular disease by lowering homocysteine levels. However, the current evidence suggests that there is no association between vitamin B12 supplementation and the risk of cardiovascular disease.

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Dementia and cognitive function

Elevated homocysteine levels may have a negative effect on the brain, and this seems to be supported by observational studies that have found correlations between low serum vitamin B12 concentrations and poor cognitive function. However, some observational studies have reported no such correlation, and randomized controlled trials have shown no improvement in cognitive function among older adults with or without dementia, mild cognitive impairment, or Alzheimer's disease when supplemented with vitamin B12 alone or in combination with folic acid, vitamin B6, or both. More evidence is needed to elucidate the effects of vitamin B12 supplementation on cognitive function and cognitive decline.

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Energy and endurance

Vitamin B12 supplementation does not appear to improve athletic performance or endurance in people who have sufficient vitamin B12 status.

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Health Risks from Excessive Vitamin B12

Vitamin B12 does not have a tolerable upper intake level because it is generally considered to be safe, even at high doses.

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Interactions with Medications

Vitamin B12 supplements may interact with medications, and some medications, including gastric acid inhibitors and metformin, may affect vitamin B12 levels.

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Vitamin B12 and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone,

especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

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